Marshmallow Robot Skins: Edible Human-Robot Interaction

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Advised by Ellen Do

Background

Social interaction research in HRI often centers a human-robot binary when studying relationship development. Social robot designs similarly target direct relation to users through speech, movement, and allusion to existing socially relational forms (animals, humans). Yet, few robot designs utilize materiality as another axis for human-robot relatability. Speculative design [1] applied to HRI offers experimental forms, functions, and design processes that, through including more-than-human perspectives, [2] can strengthen relational robot design. [3]

Problem

Expanding methods of building relationality in human-robot interaction.

Question

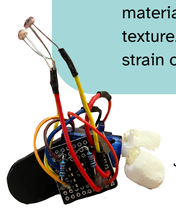
How can untraditional materials be used in robot design to create novel human-robot interactions?

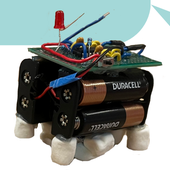
Inspiration

Emerging from the nascent field of transient robotics, [4] edible materials minimize e-waste and offer unique design characteristics and use cases for robot bodies.

Implementation

- To ensure small, playful designs the electronics were derived from analog robotics and used ATTiny85s and simple ICs to dictate movement.
- Marshmallows were chosen as the primary edible material due to their breadth in size, shape, and texture. Their light weight also minimizes weight strain on motors.

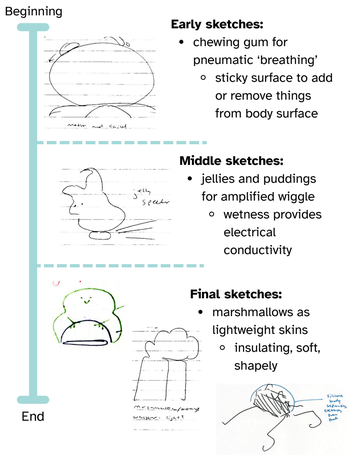




Analog robot designs adapted from David Hrynkiw & Mark Tilden's book: Junkbots, Bugbots, and Bots on Wheels

Design

- Insect-like creatures taken from analog BEAM robot designs
- Confectionary production- swirls, dollops, and blobs.
- Leveraging food mechanical properties- staleness, yield, stretch.



Final Designs



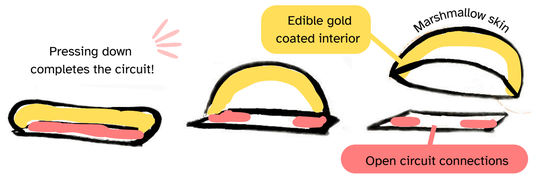


Marsnail

- Two photo-resistors act as 'eyes' that read light exposure and direct motor movement to chase a light source.
- Thick marshmallow swirls imitate a shell structure and hard pretzels act as 'legs' pulling the robot towards a light source.

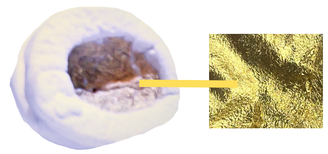
The Wiggler

- Vibrates and moves according to an analog 'nervous network.'
- Marshmallow exterior creates soft, bouncy barrier between humans and electronics. The robot's 'head' contains embedded edible circuitry which, when pressed, makes the robot glow through orifices.

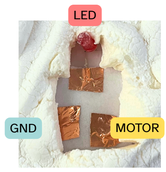


Emdedded Interaction

While marshmallows are incredible insulators due to the miniature air pockets that make up their structure, 24k edible gold retains gold's conductive properties, allowing for edible circuits to be embedded on marshmallows to create edible interfaces. The Wiggler incorporated a simple circuit using these principles: when the marshmallow 'head' was squished, its edible gold coated interior made contact with the circuit nodes thus closing the circuit, turning off the motors, and lighting an LED within the robot. The glow could be seen coming out of the robot's 'eyes' and 'mouth.'



Marshmallow head interior covered in edible gold



Open circuit nodes exposing GND, LED anode, and motor kill switch connections are secured with copper tape on a food-safe siclicone surface that separates electronics from edible materials

Reflections

- The material properties of marshmallows was a driving force in the design, production, and engagement of these robots.
- In an informal play test, participants expressed affection and delight upon seeing them wiggle around and remarked on their heightened attachment to the robots since their digestible forms gave them the ability to 'die'.
  - Additionally, texture, yield, smell, and taste worked alongside cultural connotations of marshmallows as 'cute' treats that further imbued participants interpretation of the robots' behavior.
- Edible materials in robot design offer new ways to build, envision, and interact with robots.

Future Work

- Includes formal user testing with different user groups to study appeal of edible robots.
- Design workshops where participants use similar sugary materials (gum, chocolate, gummies, etc.) to create their own robots/robot skins may shed light on affordances of other confectionary materials in robot design.
- My overall research aims to explore materiality through the lens of more-than-human design, focusing on how materials with ephemeral and evocative properties can be used to design new relations in HRI.

Sources

[1] Dunne, A., & Raby, F. (2013). Speculative Everything: Design, Fiction, and Social Dreaming. The MIT Press.
[2] Nicenboim, I., Oogjes, D., Biggs, H., & Nam, S. (2025). Decentering Through Design: Bridging Posthuman Theory with More-than-Human Design Practices. Human–Computer Interaction, 40(1–4), 195–220. https://doi.org/10.1080/07370024.2023.2283535
[3] Gemeinboeck, P., & Saunders, R. (2022). Moving beyond the mirror: Relational and performative meaning making in human–robot communication. AI & SOCIETY, 37(2), 549–563. https://doi.org/10.1007/s00146-021-01212-1
[4] Qi, Qiukai, Valentina Lo Gatto, Alexander Keller, et al. (2024) "The Rise of Transient Robotics." Device 2, no. 10. 100554. https://doi.org/10.1016/j.device.2024.100554.

Have more food for thought? Feel free to contact me at Klara.Nitsche@colorado.edu

Or check out my website to see more of my work!



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